SUPERIOR UNIVERSITY

DEPARTMENT of ENGIEERING AND TECHNOLOGY

B.s.c. information engineering technology

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TOPIC: WEB SCRAPING PROJECT

SUB-TOPIC: EXTRACTING DATA FROM Zillow.com

SUBMITTED TO: MISS KINZA FAYYAZ

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Project report

Web Scraping with Selenium and BeautifulSoup

Summary:

This report documents a web scraping script developed in Python using Selenium and BeautifulSoup to extract data from the Zillow real estate website. The script automates a browser to open the Zillow homepage, waits for the page to load, and extracts various elements such as images, text content, links, and headings. The extracted data is then saved into organized directories and text files for further use. The script includes error handling to ensure robustness and reliability. Recommendations for improvements include enhanced error handling, support for pagination, and the use of a database for efficient data storage and querying.

Introduction:

This short report provides a comprehensive overview of a web scraping script using Selenium and BeautifulSoup. The script is designed to scrape a real estate website, extracting various elements such as images, text content, links, and headings.

Methodology:

Tools and Libraries

- **Python**: The primary programming language used.
- **BeautifulSoup**: A library used for parsing HTML and XML documents.
- **Selenium**: A tool for automating web browsers, used to handle dynamic content.
- **Requests**: A library to send HTTP requests and handle web responses.

Steps

1. **Initialize the Web Driver**: Use Selenium to open the web page.
2. **Wait for the Page to Load**: Ensure all elements are fully loaded.
3. **Parse HTML Content**: Use BeautifulSoup to parse the HTML content of the page.
4. **Extract Data**: Extract images, text content, links, and headings.
5. **Save Data**: Organize and save the extracted data into files.
6. **Error Handling**: Manage potential errors during the scraping process.

Goal:

The primary goal of this report is to document the process and code for extracting and organizing data from the Zillow real estate website. The script aims to:

1. Open the Zillow homepage.
2. Wait for the page to load fully.
3. Extract and save images, text content, links, and headings.
4. Handle any errors that may occur during the process.

Conclusion and Results:

The script successfully achieves the following:

- Initializes the Chrome driver using Selenium and WebDriverManager.
- Opens the Zillow website and waits for the page to load.
- Extracts images, text content, links, and headings using BeautifulSoup.
- Saves the extracted data into organized directories and text files.
- Handles potential errors gracefully, ensuring robustness.

The extracted data is saved in the following directories:

- `images/`: Contains downloaded images.
- `text/content.txt`: Contains extracted text content.
- `links/links.txt`: Contains extracted links.
- `headings/headings.txt`: Contains extracted headings.

References:

1. **BeautifulSoup Documentation**
2. **Selenium Documentation**
3. Requests Documentation
4. **Web Scraping with Python - BeautifulSoup Tutorial**
5. **Web Scraping using Python (Beginners Tutorial)**